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# ==========================================================
# Assignment 3B: Fashion MNIST Classification using CNN
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# Step 1: Import Libraries
import tensorflow as tf
from tensorflow import keras
import numpy as np
import matplotlib.pyplot as plt

# Step 2: Load Dataset (Fashion MNIST)
(x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()

# Step 3: Preprocess Data
x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0

# Reshape to (28, 28, 1) for CNN
x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)

# Step 4: Build CNN Model
model = keras.Sequential([
    keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(64, (3,3), activation='relu'),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(128, (3,3), activation='relu'),
    keras.layers.Flatten(),

    keras.layers.Dense(128, activation='relu'),
    keras.layers.Dropout(0.25),

    keras.layers.Dense(10, activation='softmax')
])

# Step 5: Compile Model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Step 6: Train Model
history = model.fit(
    x_train, y_train,
    epochs=10,
    validation_data=(x_test, y_test),
    verbose=1
)

# Step 7: Evaluate Model
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=0)

print("\nTest Accuracy:", test_acc)
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Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx
29515/29515 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
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26421880/26421880 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
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5148/5148 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
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4422102/4422102 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarni
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 67s 35ms/step - accuracy: 0.7849 - loss: 0.5732 - val_accuracy: 0.8
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 64s 34ms/step - accuracy: 0.8617 - loss: 0.3753 - val_accuracy: 0.8
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 68s 36ms/step - accuracy: 0.8791 - loss: 0.3287 - val_accuracy: 0.8
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 64s 34ms/step - accuracy: 0.8900 - loss: 0.2993 - val_accuracy: 0.8
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 64s 34ms/step - accuracy: 0.8953 - loss: 0.2844 - val_accuracy: 0.9
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 81s 34ms/step - accuracy: 0.9024 - loss: 0.2643 - val_accuracy: 0.9
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 82s 34ms/step - accuracy: 0.9034 - loss: 0.2556 - val_accuracy: 0.9
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 81s 33ms/step - accuracy: 0.9069 - loss: 0.2455 - val_accuracy: 0.9
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 63s 34ms/step - accuracy: 0.9107 - loss: 0.2383 - val_accuracy: 0.9
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 63s 34ms/step - accuracy: 0.9132 - loss: 0.2297 - val_accuracy: 0.9

Test Accuracy: 0.9067999720573425
```